

Linux Bootcamp - Fully Corrected Version

Linux Bootcamp — Complete Corrected Version

DAY 0: Linux Foundations & Core Concepts

Core Focus: Operating system ke basic concepts aur theoretical model samajhna — koi terminal nahi kholna, sirf padhna.

- **OS Basics & Linux vs Windows** — structural differences
- **Kernel vs Shell** — core architecture relation
- **Terminal vs Shell** — interface vs interpreter separation
- **CLI vs GUI & Distributions** — command line familiarity, Linux flavors
- **File Hierarchy** — sab kuch “/” (root) se shuru hota hai, FHS standard
- **Users & Permissions** — Owner/Group/Others ke liye Read/Write/Execute
- **What is a Process?** — running program, unique PID se track hota hai
- **Package Manager** — automatic software installer (apt, yum, dnf)
- **Environment Variables** — system-wide settings (jaise PATH)
- **What is SSH?** — remote server ko securely control karne ka protocol
- **What is Systemd?** — boot aur background services ka manager
- **Web Server vs App Server** — Nginx/Apache static content, Node.js/Tomcat app logic
- **DevOps Culture** — Dev aur Ops teams ko automation ke sath jodna

Free Resources: linuxjourney.com (Getting Started section)

DAY 1: Navigation, File Ops, Viewing & Link Management

Core Focus: File hierarchy, directory structure, aur file/folder ke sath basic kaam karna — dekhna, banana, edit karna.

Navigation: `pwd, cd (~, -, ..), ls -la, ls -lh, ls -lt, history`

Directory Structure: `/etc, /var/log, /home, /usr, /bin & /sbin, /proc, /tmp, /opt, /root`

Concepts: Inodes aur File Types ka theoretical aur practical usage

Commands: `stat filename` (inode/metadata dekhna), `file filename` (file type identify karna)

File Operations: `mkdir -p, touch, cp -r, mv, rm -rf, ln, ln -s` (hard/soft links)

Viewing & Editing: cat, less, head, nano, vim

Searching: find, xargs

Free Resources: linuxjourney.com (Grasshopper section), Kunal Kushwaha — YouTube (Linux for DevOps Playlist)

DAY 2: Permissions, Ownership & User Management

Core Focus: Security, access control aur user management.

Permissions: chmod (755/644/400/600), chown, chgrp, Octal & symbolic notation (u+x, g-w, o=r), umask

DevOps Best Practice: SSH private key files ko chmod 400 se secure karna.

User Management: useradd -m, usermod -aG sudo, userdel -r, passwd, su - vs sudo, whoami, id, groups

Concept: su - poori tarah doosre user mein switch karta hai, sudo sirf temporary root access deta hai.

Free Resources: Kunal Kushwaha — YouTube (Permissions Deep-Dive), overthewire.org (Bandit Game — free hands-on practice)

DAY 3: Text Processing & Log Filtering

Core Focus: Log files se text filter karna, live data streams monitor karna.

Live Monitoring: tail -f

Filtering & Processing: grep -ri, awk, sed, cut, sort -n -r, uniq -c, tee -a, diff -u, wc -l

Free Resources: Kunal Kushwaha — YouTube (Text Utilities Video), overthewire.org (Bandit Game Levels 0-10)

DAY 4: Process & Service Management

Core Focus: Active processes aur services ko track/manage karna, runtime pe.

Process Operations: ps aux, top, htop, kill PID (graceful, try this first), kill -9 (force kill, last resort), pkill, pgrep

Background Jobs: &, jobs, bg, fg, nohup

Advanced Debugging: lsof (port/file usage), strace (system calls tracking)

Free Resources: linuxjourney.com (Processes section), freeCodeCamp — YouTube (Linux Process Management)

DAY 5: SSH Keys & Networking

Core Focus: Secure remote connectivity, port checking, network validation.

SSH: `ssh-keygen -t ed25519, ssh-copy-id, ssh -i, ~/.ssh/config, ~/.ssh/authorized_keys, ~/.ssh/known_hosts`

Networking: `ip a, ss -tulwn, ping, curl, wget, dig, nslookup (dig ka alternative), traceroute (network path debug karna), nc (netcat — port connectivity test karna)`

File Transfer: `scp, rsync`

Local DNS & Firewall: `/etc/hosts, ufw`

Free Resources: Kunal Kushwaha — YouTube (Networking & SSH Masterclass), docs.github.com (SSH setup official guide)

DAY 6: Storage & Package Management

Core Focus: Disk partitioning, storage check, software packaging.

Storage: `df -h, du -sh, lsblk, mount, umount`

Archiving & Compression: `tar -cvzf / tar -xvzf, gzip / gunzip, zip / unzip`

LVM Basics: `pvdisplay, vgdisplay, lvdisplay`

Partition Info: `fdisk -l / parted -l` — partitions ki list dekhna

Package Management: `apt update/install/remove, yum / dnf`

Free Resources: linuxjourney.com (Storage & File Systems), freeCodeCamp — YouTube (Linux Package Managers)

DAY 7: Automation & Bash Scripting

Core Focus: Tasks ko automatic schedules aur code scripts ke zariye handle karna.

Cron Jobs: `crontab -e, crontab -l, crontab -r` Syntax: minute hour day month weekday command

Bash Scripting: Shebang (`#!/bin/bash`), variables, `$1/$2, $?`, if-else, loops, functions

Free Resources: freeCodeCamp — YouTube (Bash Scripting Full Course), crontab.guru (test cron syntax live)

DAY 8: Environment Variables & Systemd Mastery

Core Focus: Global pathways mapping, system services management, final pipeline verification.

Environment Variables: export, echo \$PATH, printenv, source
~/.bashrc, aliases

Systemd Services & Logs: systemctl, journalctl, dmesg
(kernel/hardware boot messages)

Free Resources: Full Practical Revision on your Ubuntu Terminal
covering Day 1-7, overthewire.org (Higher Bandit Levels)

What Was Fixed (Round 1 + Round 2)

Round 1: 1. **cat, less, head, nano, vim** — Day 3 se Day 1 mein move kiye, kyunke ye file viewing/editing hain, text filtering nahi 2. **history** — Day 1 mein add kiya 3. **printenv** — Day 8 mein add kiya 4. **Day 0 expand kiya** — 9 naye concepts add kiye

Round 2 (Deeper Check): 5. **whoami, id, groups** — Day 2 mein add kiye (basic user info commands) 6. **stat, file** — Day 1 mein add kiye (Inodes/File Types concept ko command ke sath match kiya) 7. **Plain kill PID, pgrep** — Day 4 mein add kiye (graceful kill ka best practice — hamesha -9 se pehle try karni chahiye) 8. **traceroute, nc** — Day 5 mein add kiye (network troubleshooting) 9. **gzip/gunzip, zip/unzip** — Day 6 mein add kiye (compression ke alternatives, sirf tar tha) 10. **fdisk -l/parted -l** — Day 6 mein add kiya (partition listing) 11. **dmesg** — Day 8 mein add kiya (kernel/hardware messages)